Literature Survey

S.No	Title	Authors	Algorithm Used	Accuracy	Limitations	Future Enhancement
1.	A Hybrid Feature Fusion Deep Learning Framework for Leukemia Cancer Detection in Microscopic Blood Sample Using Gated Recurrent Unit and Uncertainty Quantification	Maksuda Akter, Rabea Khatun, and Md Manowarul Islam	MobileNetV2-GRU, InceptionV3-GRU, EfficientNetB3-GRU	98.08%	The framework relies on small base datasets of only 108 and 260 original images, relying heavily on artificial data augmentation.	Future research will focus on expanding the model to identify and classify specific leukemia subtypes.
2.	Acute Lymphoblastic Leukemia Detection Using Hypercomplex-Valued Convolutional Neural Networks	Guilherme Vieira and Marcos Eduardo Valle	Hypercomplex-Valued Convolutional Neural Network (HvCNN),Real-Valued Convolutional Neural Network (RvCNN)	93.8%	The computational experiments were limited strictly to the ALL-IDB2 dataset (consisting of only 260 segmented microscopic cell images), which is a relatively small dataset.	Developing native hypercomplex-valued pooling, weight initialization, and batch normalization layers rather than relying on real-valued emulation.
3.	A study on deep feature extraction to detect and classify Acute Lymphoblastic Leukemia (ALL)	Sabit Ahamed Preanto, Dr. Md. Taimur Ahad, Yousuf Rayhan Emon	ResNet101 Inception V3 DenseNet121 MobileNetV2	87.0%	Certain classifiers like Naïve Bayes performed poorly across almost all feature extraction setups, dropping as low as 41.6% accuracy when paired with Inception V3.	Training the models on larger, more diverse datasets to improve generalization and robust clinical viability.
